K L UNIVERSITY

FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

On

Automated Household Inventory & Intelligent Consumption Analytics System (HI-CAS)

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CERTIFICATE

This is to certify that the project based laboratory report entitled "Automated Household Inventory & Intelligent Consumption Analytics System (HI-CAS)" submitted by **Ch. Anil, J. Rohan , T. Siddhardha, vasantha vani** bearing Regd. No.2500030315, 2500030366, 2500030478, 2500030615 to the **Department of COMPUTER SICENCE ENGINEERING , KL University** in partial fulfillment of the requirements for the completion of a project based Laboratory in "Python Programming (25SC2007)" course in I B Tech II Semester, is a Bonafede record of the work carried out by him/her under my supervision during the academic year 2025 – 2026.

PROJECT SUPERVISOR

Mr. K. SAI SANDEEP

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Finally, we are pleased to acknowledge our indebtedness to all those who devoted themselves directly or indirectly to making this project a success.

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ABSTRACT

The Automated Household Inventory & Intelligent Consumption Analytics System (HI-CAS) is a smart, data-driven solution designed to modernize the management of household resources through automation, real-time tracking, and predictive analytics. The system integrates Internet of Things (IoT) sensors, barcode/RFID scanning, and mobile or web-based interfaces to continuously monitor the availability, usage, and lifecycle of household items such as groceries, utilities, and consumables.

INDEX

S.NO	TITLE	PAGE NO
1	Introduction	6
2	Aim of the Project	7
2.1	Advantages & Disadvantages	8
2.2	Future Implementation	9
3	Software & Hardware Details	10
4	Algorithm for each module	12
5	Flow Chart	13
6	Implementation	14-16
7	Integration and System Testing	17
8	Conclusion	18

INTRODUCTION

The Automated Household Inventory and Consumption Analytics System is a smart solution designed to simplify household resource management. By integrating automation with data analytics, the system eliminates manual tracking and provides real-time insights into consumption patterns. It ensures that families can efficiently monitor supplies, reduce waste, and optimize budgets through intelligent recommendations and predictive analysis.

AIM

The aim of this project is to design and develop an Automated Household Inventory and Consumption Analytics System that minimizes manual effort by automatically tracking household items, analyzing consumption trends, and generating intelligent recommendations. The system seeks to enhance convenience, reduce waste, and promote sustainable living through real-time monitoring and predictive analytics.

ADVANTAGES

- **Efficient Inventory Management**: Keeps a structured record of items, quantities, categories, expiry dates, and prices, reducing manual effort.

- **Reduced Waste**: Tracks expiry dates and usage patterns to minimize spoilage and unused items.

- **Cost Optimization:** Provides monthly analytics and category-wise spending insights to help households plan budgets better**.**

DISADVANTAGES

- **Data Accuracy Dependence:** The system relies heavily on correct input; wrong quantities, dates, or categories can lead to inaccurate analytics**.**

- **Initial Setup Effort:** Users must manually enter all grocery items, categories, expiry dates, and prices, which can be time-consuming.

- **Resource Usage:** Running analytics and visualizations may require moderate computing resources, which could be challenging for very low-end systems.

FUTURE IMPLEMENTATION

- **Mobile App Integration**: Develop a mobile interface for real-time inventory updates, notifications, and easy accessibility.

- **Barcode/QR Code Scanning:** Allow users to quickly add or update items by scanning product codes instead of manual entry.

SYSTEM REQUIREMENTS

➢ SOFTWARE REQUIREMENTS:

The major software requirements of the project are as follows:
Language: Python.
Operating system**:** Windows 10 or more.
Technical requirements: Monitor, CPU, Keyboard, Mouse, etc...

➢ HARDWARE REQUIREMENTS:

The hardware requirements that map towards the software are as follows:

RAM: 8 GB.

Processor: i5/i7.

ALGORITHM

1. Start System

2. User Login / Authentication

- Verify credentials and load household data.

3. Automated Inventory Update

- Use sensors, barcode scanning, or purchase records to auto-update items.

4. Consumption Tracking

- System automatically detects usage and reduces item quantities.

5. Analytics Module

- Analyze consumption trends (daily, weekly, monthly).
- Predict shortages and detect overstock.
- Identify wastage (expired items).

6. Decision Support

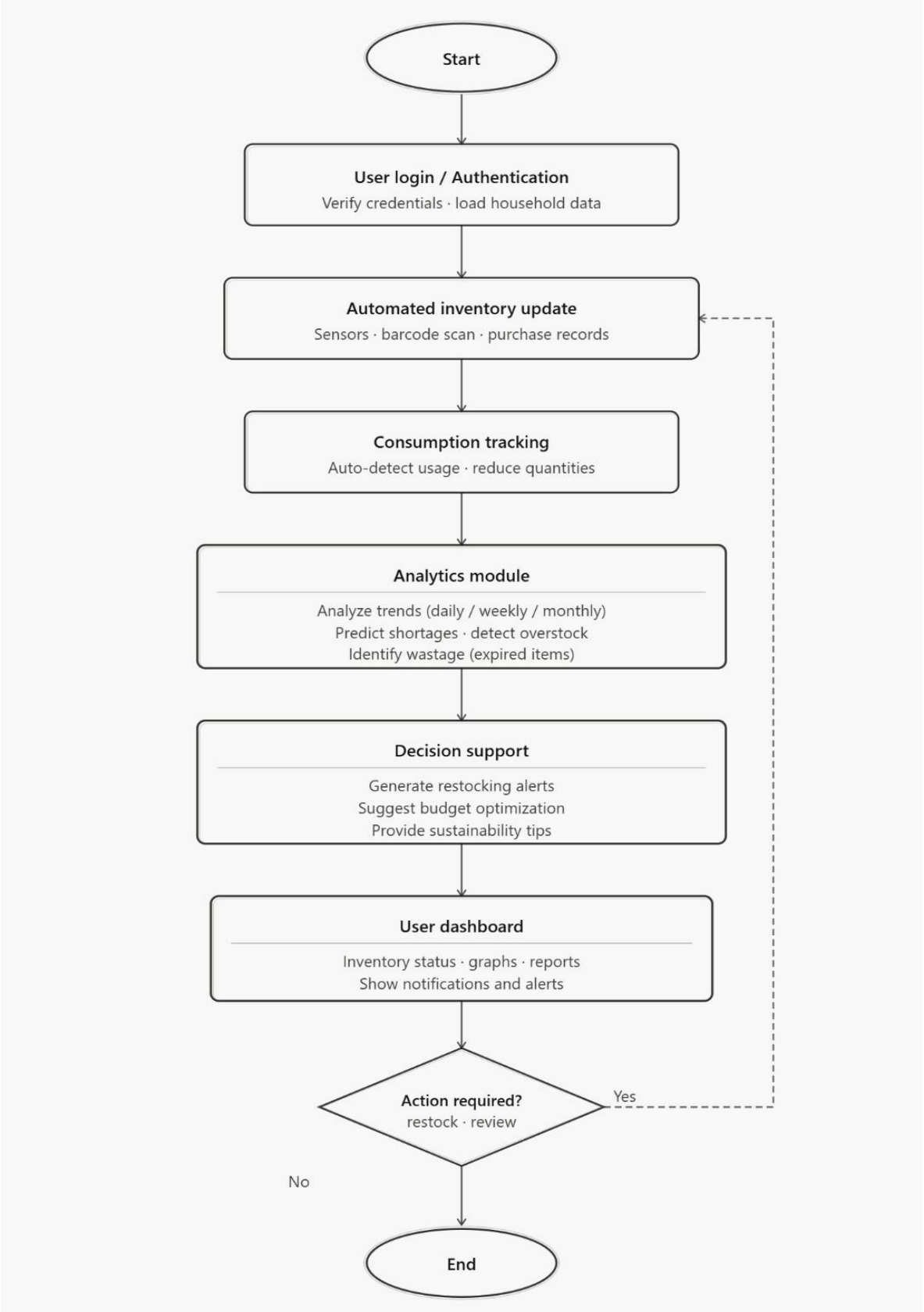
- Generate restocking alerts.
- Suggest budget optimization.
- Provide sustainability tips.

7. User Dashboard

- Display inventory status, graphs, and reports.
- Show notifications and alerts

8. End

FLOWCHART


Start
User login / Authentication
Verify credentials · load household data
Automated inventory update
Sensors · barcode scan · purchase records
Consumption tracking
Auto-detect usage · reduce quantities
Analytics module
Analyze trends (daily / weekly / monthly)
Predict shortages · detect overstock
Identify wastage (expired items)
Decision support
Generate restocking alerts
Suggest budget optimization
Provide sustainability tips
User dashboard
Inventory status · graphs · reports
Show notifications and alerts
Action required?
restock · review
Yes
No
End

IMPLEMENTATION

```
import datetime

class HICAS:
    def __init__(self):
        self.users = {"admin": "password123"}
        self.inventory = {}
        self.log = []

    def login(self):
        u = input("Username: ")
        p = input("Password: ")
        return u in self.users and self.users[u] == p

    def add_item(self):
        item = input("Item: ")
        qty = int(input("Quantity: "))
        exp = input("Expiry (YYYY-MM-DD or blank): ")
        expiry = datetime.datetime.strptime(exp, "%Y-%m-%d").date() if exp else None
        self.inventory[item] = {"qty": self.inventory.get(item, {}).get("qty", 0) + qty, "exp": expiry}

    def use_item(self):
        item = input("Consume item: ")
        qty = int(input("Quantity: "))
        if item in self.inventory and self.inventory[item]["qty"] >= qty:
            self.inventory[item]["qty"] -= qty
            self.log.append((item, qty, datetime.datetime.now()))
        else:
            print("Not enough stock!")

    def dashboard(self):
        print("\n--- Dashboard ---")
        for i, d in self.inventory.items():
            print(f"{i}: {d['qty']} (Expiry: {d['exp']})")
        today = datetime.date.today()
        for i, d in self.inventory.items():
            if d["qty"] < 2:
                print(f" Low stock: {i}")
            if d["exp"] and d["exp"] < today:
                print(f" Expired: {i}")
```

```
if __name__ == "__main__":
    sys = HICAS()
    if sys.login():
        while True:
            print("\n1.Add  2.Use  3.Dashboard  4.Exit")
            c = input("Choice: ")
            if c == "1": sys.add_item()
            elif c == "2": sys.use_item()
            elif c == "3": sys.dashboard()
            elif c == "4": break
```

INTEGRATION AND SYSTEM TESTING OUTPUTS

Screen Shots:

main.py

```python
import datetime

class HICAS:
    def __init__(self):
        self.users = {"admin": "password123"}
        self.inventory = {}
        self.log = []

    def login(self):
        u = input("Username: ")
        p = input("Password: ")
        return u in self.users and self.users[u] == p

    def add_item(self):
        item = input("Item: ")
        qty = int(input("Quantity: "))
        exp = input("Expiry (YYYY-MM-DD or blank): ")
        expiry = datetime.datetime.strptime(exp, "%Y-%m-%d").date() if exp else None
        self.inventory[item] = {"qty": self.inventory.get(item, {}).get("qty", 0) + qty, "exp": expiry}

    def use_item(self):
        item = input("Consume item: ")
```

input

```
Username: admin
Password: password123

1.Add  2.Use  3.Dashboard  4.Exit
Enter Choice: 1
Item: rice
Quantity: 22
Expiry (YYYY-MM-DD or blank): 2026-04-22

1.Add  2.Use  3.Dashboard  4.Exit
Enter Choice: 1
Item: dal
Quantity: 20
```

```
Username: admin
Password: password123

1.Add  2.Use  3.Dashboard  4.Exit
Enter Choice: 1
Item: rice
Quantity: 22
Expiry (YYYY-MM-DD or blank): 2026-04-22

1.Add  2.Use  3.Dashboard  4.Exit
Enter Choice: 1
Item: dal
Quantity: 20
Expiry (YYYY-MM-DD or blank): 2026-04-20

1.Add  2.Use  3.Dashboard  4.Exit
Enter Choice: 2
Consume item: rice
Quantity: 20

1.Add  2.Use  3.Dashboard  4.Exit
Enter Choice: 2
Consume item: rice
Quantity: 25
Not enough stock!

1.Add  2.Use  3.Dashboard  4.Exit
Enter Choice: 3

--- Dashboard ---
rice: 2 (Expiry: 2026-04-22)
dal: 20 (Expiry: 2026-04-20)

1.Add  2.Use  3.Dashboard  4.Exit
Enter Choice: 4
```

CONCLUSION

The Automated Household Inventory and Consumption Analytics System transforms household management into a seamless, data-driven process. By automating inventory updates and consumption tracking, it reduces manual effort and ensures accuracy. The system empowers families with timely alerts, predictive insights, and intelligent recommendations, ultimately saving time, reducing costs, and promoting sustainable living.